Disproof of Conjecture 00000000317

The Last Math Competition

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Abstract

Conjecture 00000000317 is false. It asserts a conjunction of two clauses about Freiman's constant c_F. The second clause, that the digit 1 occurs exactly 8 times among the first 20 partial quotients of c_F, is refuted by an exact computation: the digit 1 occurs exactly 5 times among $a_0, \dots, a_{19}$, and exactly 6 times under the shifted indexing $a_1, \dots, a_{20}$. Since the conjunction requires both clauses, it fails whatever the first clause says. The computation is formalised in Lean 4 with Mathlib in the file `Disproof.lean`. The first clause ($c_F \in L$ and $c_F \notin M$) is also false, as Section 3.3 explains, but the formal refutation does not need this.

1 The conjecture as filed

The conjecture file states the following (English part; the Chinese part of the same file is a literal translation of this text and is not used here). The mathematical symbols are typeset in math mode; the wording is unchanged.

> Definition: Freiman's constant c_F is the largest value for which $(2\sqrt{2}, c_F) \cap M = \emptyset$. Conjecture: c_F itself belongs to L but not to M (an endpoint asymmetry of the spectra), and among the first 20 partial quotients of c_F the digit 1 appears exactly 8 times.

The conjecture is therefore a conjunction of two clauses:

(i) $c_F \in L$ and $c_F \notin M$;

(ii) the digit 1 occurs exactly 8 times among the first 20 partial quotients of c_F.

Clause (ii) is false by exact computation (Section 3). Clause (i) is false as well (Section 3.3). Either one suffices to refute the conjunction; the Lean proof uses (ii).

2 Precise reading

2.1 Freiman's constant

Let L and M be the Lagrange and Markov spectra (subsets of $\mathbb{R}$; see Cusick and Flahive [2]). Freiman's constant is the number c_F of the definition above. Freiman's 1975 work [1] gives its closed form

$$c_F \;=\; \frac{2221564096 + 283748\sqrt{462}}{491993569} \;\approx\; 4.5278295661.$$

This closed form is taken as the definition of c_F in the formal file. The spectral characterisation (the largest c with $(2\sqrt{2}, c) \cap M = \emptyset$) is not reproved here. Since $\sqrt{462}$ is irrational and the coefficient 283748 is nonzero, c_F is irrational.

2.2 Regular continued fraction via the Gauss map

For real x write $\{x\} = x - \lfloor x \rfloor$. Define the complete quotients and partial quotients by

$$x_0 = x, \qquad x_{n+1} = \frac{1}{\{x_n\}}, \qquad a_n = \lfloor x_n \rfloor.$$

For irrational x this gives the regular continued fraction $x = [a_0; a_1, a_2, \dots]$. For c_F we have $x_0 = c_F$ and $a_0 = 4$.

2.3 Indexing of "the first 20 partial quotients"

The phrase admits three natural readings. The conjecture is refuted under all three.

(W1) $a_0, \dots, a_{19}$: twenty terms counting the integer part a_0 as the first.

(W2) $a_1, \dots, a_{20}$: twenty terms, with a_0 not counted as a partial quotient.

(W3) $a_0, \dots, a_{20}$: twenty-one terms, included for completeness.

The primary reading is (W1). The Lean file proves (W1) and (W2) explicitly. Reading (W3) is not a separate theorem in the file; its count is 6, since $a_0 = 4$ is not a 1 and $a_{20} = 1$.

3 The computation

3.1 Method

Every complete quotient is kept in the exact form $x_n = (p_n + r_n\sqrt{462})/s_n$ with integers p_n, r_n and $s_n > 0$. Put $a = a_n = \lfloor x_n \rfloor$ and $q = \sqrt{462}$. Then

$$x_{n+1} = \frac{1}{x_n - a} = \frac{s_n\big((p_n - as_n) - r_n q\big)}{(p_n - as_n)^2 - 462\, r_n^2}.$$

Multiplying numerator and denominator by a common factor g_n (chosen to cancel the gcd of the new integers) gives the next triple $(p_{n+1}, r_{n+1}, s_{n+1})$, with $s_{n+1} > 0$. The starting triple is $(p_0, r_0, s_0) = (2221564096,\ 283748,\ 491993569)$.

Each floor is certified by an inequality: for $s > 0$, $\lfloor (p + rq)/s \rfloor = a$ holds iff $as \le p + rq < (a+1)s$. The bounds used are

$$\frac{21494185260204}{10^{12}} < \sqrt{462} < \frac{21494185260205}{10^{12}},$$

which hold because $21494185260204^2 < 462 \cdot 10^{24} < 21494185260205^2$. The script `check.py` verifies that these bounds imply every one of the 21 floor inequalities.

3.2 Result

Table 1 lists the 21 complete quotients $x_0, \ldots, x_{20}$ with their integer triples and partial quotients, as produced by `check.py`. The last column is a floating-point approximation for orientation only; the proof uses only the exact triples.

The partial quotients are

$$a_0, \ldots, a_{20} = 4,\ 1,\ 1,\ 8,\ 2,\ 14,\ 2,\ 2,\ 1,\ 12,\ 117,\ 1,\ 1,\ 7,\ 16,\ 35,\ 6,\ 2,\ 7,\ 2,\ 1.$$

The digit 1 occurs at indices $n = 1, 2, 8, 11, 12$ among $a_0, \ldots, a_{19}$, and additionally at $n = 20$ for the shifted reading. Therefore:

- (W1) $a_0, \ldots, a_{19}$: the digit 1 occurs **5** times;
- (W2) $a_1, \ldots, a_{20}$: the digit 1 occurs **6** times;
- (W3) $a_0, \ldots, a_{20}$: the digit 1 occurs **6** times.

n	p_n	r_n	s_n	a_n	x_n (approx.)
0	2221564096	283748	491993569	4	4.5278295662
1	63397455	−70937	32658252	1	1.8945509386
2	122956812	283748	115446937	1	1.1178793256
3	7509875	−283748	166321	8	8.4832517889
4	6179307	283748	5933481	2	2.0693146366
5	5687655	283748	816983	14	14.4269673656
6	5750107	283748	5059153	2	2.3420993750
7	4368199	283748	3580799	2	2.9231272348
8	2793399	283748	8208753	1	1.0832742902
9	2707677	141874	479422	12	12.0085082445
10	6090774	283748	103713	117	117.5330583361
11	6043647	283748	6472703	1	1.8759672859
12	107264	70937	1429576	1	1.1415951442
13	5289248	283748	1612511	7	7.0623890809
14	5998329	283748	754737	16	16.0284457754
15	6077463	283748	346367	35	35.1546050265
16	3022691	141874	938786	6	6.4680950074
17	5220050	283748	5298359	2	2.1363184486
18	1344167	70937	391084	7	7.3357642343
19	5573684	283748	3919247	2	2.9782802868
20	1132405	141874	4091042	1	1.0222019328

Table 1: Complete quotients $x_n = (p_n + r_n\sqrt{462})/s_n$ and partial quotients $a_n = \lfloor x_n \rfloor$ of c_F, $0 \le n \le 20$.

In no reading does the digit 1 occur 8 times, so clause (ii) is false.

As an independent check, `check.py` compares the 21 quotients with SymPy's `continued_fraction_iterator` applied to the closed form and finds exact agreement. For orientation only, the same SymPy run gives $a_{21} = 45$, $a_{22} = 24$, $a_{23} = 3$; these values are not used in the disproof.

3.3 Clause (i) is also false

The first clause, $c_F \in L$ and $c_F \notin M$, is false by the classical inclusion $L \subseteq M$ (Cusick and Flahive [2]): $c_F \in L$ would force $c_F \in M$. An independent argument uses only that M is closed and that c_F is maximal. If $c_F \notin M$, closedness gives $\varepsilon > 0$ with $(c_F-\varepsilon, c_F+\varepsilon)\cap M = \emptyset$, hence $(2\sqrt{2}, c_F+\varepsilon)\cap M = \emptyset$, contradicting the maximality of c_F. So $c_F \in M$. This clause is not formalised in Lean; the main theorem treats it as an arbitrary proposition.

4 Lean formalisation

The file `Disproof.lean` lives in the submission folder and uses namespace `Freiman`. It was checked with Lean 4.33.1 and Mathlib v4.33.1. The listings below are copied from the file; Unicode symbols are rendered in math mode by the listing escape, and proofs are omitted where noted. The compile record is in `MAPPING.md` in the submission folder.

4.1 Definitions

```
/-- Freiman's constant `c_F`, the right end point of the maximal gap `(2
    √2, c_F)` of the
Markov spectrum, in its standard closed form (Freiman 1975):
`c_F = (2221564096 + 283748·√462) / 491993569 ≈ 4.5278295661...`. -/
noncomputable def freimanConstant : ℝ :=
  (2221564096 + 283748 * Real.sqrt 462) / 491993569

/-- The Gauss-map orbit of a real number `x`: `cfState x 0 = x` and
`cfState x (n+1) = 1 / fract (cfState x n)`. This is the standard recursion
     producing the
regular continued fraction of `x`: `cfState x n` is the `n`-th complete
    quotient. -/
noncomputable def cfState (x : ℝ) : ℕ → ℝ
  | 0 => x
  | n + 1 => 1 / Int.fract (cfState x n)

/-- The `n`-th partial quotient `a_n` of the regular continued fraction
`x = [a_0; a_1, a_2, ...]`, defined as the integer part of the `n`-th
    complete quotient. -/
noncomputable def pq (x : ℝ) (n : ℕ) : ℤ := ⌊cfState x n⌋
```

4.2 Rational bracketing of $\sqrt{462}$

The two bounds are proved with `Real.lt_sqrt` and `Real.sqrt_lt'` and `norm_num`. The bounds correspond to the integer inequalities in Section 3.1, scaled by 10^{12}.

```
theorem sq_sqrt462 : Real.sqrt 462 ^ 2 = 462 := Real.sq_sqrt (by norm_num)

theorem sqrt462_gt : (5373546315051 : ℝ) / 250000000000 < Real.sqrt 462 :=
  (Real.lt_sqrt (by norm_num)).2 (by norm_num)

theorem sqrt462_lt : Real.sqrt 462 < (4298837052041 : ℝ) / 200000000000 :=
  (Real.sqrt_lt' (by norm_num)).2 (by norm_num)
```

4.3 Generic step lemmas

`floor_of_bounds` turns an inequality $as \le p + rq < (a+1)s$ into a floor identity. The lemma `gauss_step` performs one Gauss-map step on triples of the form $(p + r\sqrt{462})/s$. Its proof rationalises $1/(x-a)$ and cancels the factor g; it uses `field_simp`, `div_eq_div_iff` and `linear_combination` (see the file for the tactic script).

```
/-- Floor of a number of the form `(p + r√462)/s`, given rational
    bracketing. -/
theorem floor_of_bounds (p r s : ℝ) (a : ℤ) (hs : 0 < s)
    (h₁ : (a : ℝ) * s ≤ p + r * Real.sqrt 462)
    (h₂ : p + r * Real.sqrt 462 < ((a : ℝ) + 1) * s) :
    ⌊(p + r * Real.sqrt 462) / s⌋ = a := by
  rw [Int.floor_eq_iff, le_div_iff₀ hs, div_lt_iff₀ hs]
  exact ⟨h₁, h₂⟩

/-- One Gauss-map step. If the `n`-th complete quotient is `(p + r√462)/s`
     with integer part
`a`, then the next one is `(P + R√462)/S`, where `g·P = s(p - a s)`, `g·R =
     -s r`,
`g·S = (p - a s)^2 - 462 r^2` for some non-zero common factor `g` (used to
    cancel a gcd). -/
theorem gauss_step (x : ℝ) (n : ℕ) (p r s P R S g : ℝ) (a : ℤ)
    (hx : cfState x n = (p + r * Real.sqrt 462) / s) (hs : s ≠ 0)
    (ha : ⌊cfState x n⌋ = a) (hg : g ≠ 0) (hS : S ≠ 0)
    (hP : g * P = s * (p - a * s)) (hR : g * R = -(s * r))
    (hSS : g * S = (p - a * s) ^ 2 - 462 * r ^ 2) :
    cfState x (n + 1) = (P + R * Real.sqrt 462) / S := by
```

4.4 The orbit of c_F

The file contains `state_0`, …, `state_20` and `pq_0`, …, `pq_20`. Each `state_(n+1)` is an instance of `gauss_step`, with the data (p_n, r_n, s_n) of Table 1, the cancellation factor g_n, and the new triple $(p_{n+1}, r_{n+1}, s_{n+1})$. Each `pq_n` is proved by `floor_of_bounds` with the two side goals closed by `linarith` from the bracketing lemmas. Three representative declarations are shown:

```
theorem state_0 : cfState freimanConstant 0 = (2221564096 + 283748 * Real.
    sqrt 462) / 491993569 := rfl

theorem pq_0 : pq freimanConstant 0 = 4 := by
  unfold pq
  rw [state_0]
  have hL := sqrt462_gt
```

```
  have hU := sqrt462_lt
  exact floor_of_bounds _ _ _ _ (by norm_num) (by push_cast; linarith) (by
    push_cast; linarith)

theorem state_1 : cfState freimanConstant 1 = (63397455 + (-70937) * Real.
    sqrt 462) / 32658252 :=
  gauss_step freimanConstant 0 _ _ _ _ _ _ 1967974276 _ state_0 (by
    norm_num) pq_0
    (by norm_num) (by norm_num) (by norm_num) (by norm_num) (by norm_num)
```

4.5 Counting the digit 1

```
/-- Among the first 20 partial quotients ‘a_0, ..., a_19‘ of Freiman’s
    constant, the digit 1
occurs exactly 5 times (at indices 1, 2, 8, 11, 12). -/
theorem count_ones_first20 :
    ((Finset.range 20).filter (fun n => pq freimanConstant n = 1)).card = 5
     := by
  simp only [Finset.card_filter, Finset.sum_range_succ, Finset.
    sum_range_zero,
    pq_0, pq_1, pq_2, pq_3, pq_4, pq_5, pq_6, pq_7, pq_8, pq_9, pq_10,
    pq_11, pq_12, pq_13, pq_14, pq_15, pq_16, pq_17, pq_18, pq_19]
  norm_num

/-- Variant indexing: among ‘a_1, ..., a_20‘ the digit 1 occurs exactly 6
    times. -/
theorem count_ones_1_to_20 :
    ((Finset.range 20).filter (fun n => pq freimanConstant (n + 1) = 1)).
    card = 6 := by
  simp only [Finset.card_filter, Finset.sum_range_succ, Finset.
    sum_range_zero]
  norm_num [pq_1, pq_2, pq_3, pq_4, pq_5, pq_6, pq_7, pq_8, pq_9, pq_10,
    pq_11, pq_12, pq_13, pq_14, pq_15, pq_16, pq_17, pq_18, pq_19, pq_20]
```

4.6 Main theorems

The first clause is abstracted as an arbitrary proposition `InLNotM`. Both theorems hold for every such proposition, so the refutation does not depend on how clause (i) would be formalised.

```
/-- **Conjecture 00000000317 is false.** The first clause ("‘c_F ∈ L‘ and ‘
    c_F ∉ M‘") is
abstracted as an arbitrary proposition ‘InLNotM‘: the refutation of the
    conjunction does not
depend on it, because the second clause ("the digit 1 appears exactly 8
    times among the first
```

```
20 partial quotients of ‘c_F‘") is false -- the digit 1 appears exactly 5
    times. -/
theorem conjecture_00000000317_fails (InLNotM : Prop) :
    ¬ (InLNotM ∧ ((Finset.range 20).filter (fun n => pq freimanConstant n
    = 1)).card = 8) := by
  rintro ⟨-, h⟩
  rw [count_ones_first20] at h
  norm_num at h

/-- Same conclusion if "the first 20 partial quotients" is read as ‘a_1,
    ..., a_20‘
(the digit 1 then appears exactly 6 times, still not 8). -/
theorem conjecture_00000000317_fails' (InLNotM : Prop) :
    ¬ (InLNotM ∧ ((Finset.range 20).filter (fun n => pq freimanConstant (n
     + 1) = 1)).card = 8) := by
  rintro ⟨-, h⟩
  rw [count_ones_1_to_20] at h
  norm_num at h
```

4.7 Clause-by-clause mapping

Clause	Lean object	Status
Freiman's constant c_F	`freimanConstant` (closed form as definition)	The spectral characterisation is not formalised.
Partial quotients of c_F	`cfState`, `pq` (Gauss-map recursion)	Self-contained; not via Mathlib's continued fraction API.
First 20 partial quotients, reading (W1)	`Finset.range 20` with `pq freimanConstant n`	Count 5: `count_ones_first20`.
First 20 partial quotients, reading (W2)	`Finset.range 20` with `pq freimanConstant (n + 1)`	Count 6: `count_ones_1_to_20`.
Digit 1 occurs exactly 8 times	`card` of the filter $= 8$	Refuted by the two counts above.
$c_F \in L$ and $c_F \notin M$	Abstract `InLNotM : Prop`	Not formalised; mathematically false (Section 3.3).
Conjunction	`conjecture_00000000317_fails`	Main theorem, reading (W1).
Conjunction, shifted reading	`conjecture_00000000317_fails'`	Variant theorem, reading (W2).

Table 2: Correspondence between the conjecture and the Lean declarations in `Disproof.lean`.

4.8 Axioms

The four `#print axioms` commands in the file report, for each of `count_ones_first20`, `count_ones_1_to_20`, `conjecture_00000000317_fails` and `conjecture_00000000317_fails'`, the axioms `[propext, Classical.choice, Quot.sound]`. No use of `sorryAx` is involved.

```
#print axioms Freiman.count_ones_first20
#print axioms Freiman.count_ones_1_to_20
#print axioms Freiman.conjecture_00000000317_fails
#print axioms Freiman.conjecture_00000000317_fails'
```

5 Reproduction

All commands are run from the submission folder.

1. Lean check (requires the Lean toolchain and the Mathlib cache):

   ```
   cd lean && lake exe cache get && lake build
   ```

   Expected: the build finishes with no errors, and the `#print axioms` lines show only `propext`, `Classical.choice` and `Quot.sound`.

2. Exact computation (requires `sympy`):

   ```
   python3 check.py
   ```

   Expected output: the partial quotients $4, 1, 1, 8, 2, 14, 2, 2, 1, 12, 117, 1, 1, 7, 16, 35, 6, 2, 7, 2, 1$, the counts 5 (for $a_0, \ldots, a_{19}$) and 6 (for $a_1, \ldots, a_{20}$), and the SymPy agreement line. Note that `check.py` also regenerates `Disproof.lean`.

3. Build this report:

   ```
   tectonic report.tex
   ```

References

[1] G. A. Freiman, *Diophantine Approximation and the Geometry of Numbers (Markov's Spectrum)*, Kalinin State University, 1975.

[2] T. W. Cusick and M. E. Flahive, *The Markoff and Lagrange Spectra*, Mathematical Surveys and Monographs 30, American Mathematical Society, 1989.